

By writing (6) as

$$\mathbf{a}(t) = \frac{ds}{dt} \frac{d\mathbf{T}}{dt} + \frac{d^2s}{dt^2} \mathbf{T},$$

we note that the so-called scalar acceleration d^2s/dt^2 , referred to in the last remark, is now seen to be the tangential component of the acceleration a_T .

9.3 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1 and 2, for the given position function, find the unit tangent.

1. $\mathbf{r}(t) = (t \cos t - \sin t)\mathbf{i} + (t \sin t + \cos t)\mathbf{j} + t^2\mathbf{k}, t > 0$
2. $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + \sqrt{2}e^t \mathbf{k}$
3. Use the procedure outlined in Example 2 to find $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, and κ for motion on a general circular helix that is described by $\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j} + ct \mathbf{k}$.
4. Use the procedure outlined in Example 2 to show on the twisted cubic of Example 4 that at $t = 1$:

$$\mathbf{T} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k}), \quad \mathbf{N} = -\frac{1}{\sqrt{2}}(\mathbf{i} - \mathbf{k}),$$
$$\mathbf{B} = -\frac{1}{\sqrt{6}}(-\mathbf{i} + 2\mathbf{j} - \mathbf{k}), \quad \kappa = \frac{\sqrt{2}}{3}.$$

In Problems 5 and 6, find an equation of the osculating plane to the given space curve at the point that corresponds to the indicated value of t .

5. The circular helix of Example 2; $t = \pi/4$
6. The twisted cubic of Example 4; $t = 1$

In Problems 7–16, $\mathbf{r}(t)$ is the position vector of a moving particle. Find the tangential and normal components of the acceleration at any t .

7. $\mathbf{r}(t) = \mathbf{i} + t\mathbf{j} + t^2\mathbf{k}$
8. $\mathbf{r}(t) = 3 \cos t\mathbf{i} + 2 \sin t\mathbf{j} + t\mathbf{k}$
9. $\mathbf{r}(t) = t^2\mathbf{i} + (t^2 - 1)\mathbf{j} + 2t^2\mathbf{k}$
10. $\mathbf{r}(t) = t^2\mathbf{i} - t^3\mathbf{j} + t^4\mathbf{k}$
11. $\mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j}$
12. $\mathbf{r}(t) = \tan^{-1} t\mathbf{i} + \frac{1}{2} \ln(1 + t^2)\mathbf{j}$
13. $\mathbf{r}(t) = 5 \cos t\mathbf{i} + 5 \sin t\mathbf{j}$
14. $\mathbf{r}(t) = \cosh t\mathbf{i} + \sinh t\mathbf{j}$
15. $\mathbf{r}(t) = e^{-t}(\mathbf{i} + \mathbf{j} + \mathbf{k})$
16. $\mathbf{r}(t) = t\mathbf{i} + (2t - 1)\mathbf{j} + (4t + 2)\mathbf{k}$
17. Find the curvature of an elliptical helix that is described by $\mathbf{r}(t) = a \cos t\mathbf{i} + b \sin t\mathbf{j} + ct\mathbf{k}$, $a > 0$, $b > 0$, $c > 0$.
18.
 - (a) Find the curvature of an elliptical orbit that is described by $\mathbf{r}(t) = a \cos t\mathbf{i} + b \sin t\mathbf{j} + c\mathbf{k}$, $a > 0$, $b > 0$, $c > 0$.
 - (b) Show that when $a = b$, the curvature of a circular orbit is the constant $\kappa = 1/a$.
19. Show that the curvature of a straight line is the constant $\kappa = 0$.
[Hint: Use (2) in Section 7.5.]
20. Find the curvature at $t = \pi$ of the cycloid that is described by

$$\mathbf{r}(t) = a(t - \sin t)\mathbf{i} + a(1 - \cos t)\mathbf{j}, \quad a > 0.$$
21. Let C be a plane curve traced by $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g have second derivatives. Show that the curvature at a point is given by

$$\kappa = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{([f'(t)]^2 + [g'(t)]^2)^{3/2}}.$$
22. Show that if $y = F(x)$, the formula for κ in Problem 21 reduces to

$$\kappa = \frac{|F''(x)|}{[1 + (F'(x))^2]^{3/2}}.$$

In Problems 23 and 24, use the result of Problem 22 to find the curvature and radius of curvature of the curve at the indicated points. Decide at which point the curve is “sharper.”

23. $y = x^2$; $(0, 0)$, $(1, 1)$

24. $y = x^3$; $(-1, -1)$, $(\frac{1}{2}, \frac{1}{8})$

Discussion Problems

25. Discuss the curvature near a point of inflection of $y = F(x)$.

26. Show that $\|\mathbf{a}(t)\|^2 = a_N^2 + a_T^2$.

*Literally, this means the “kissing” plane.